# Abdelilah Elgallati

**Data Scientist | AI / Machine Learning Engineer**

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Recent double Master's graduate in Data Science and Decision Support, combining strong **data science and modeling** skills with hands-on experience in **AI engineering**. Comfortable across the full pipeline, from statistical analysis, feature engineering, and model evaluation to designing and deploying production AI systems such as LLM-powered RAG services and hybrid recommendation engines. Proficient in Python, PyTorch, and Scikit-learn, with practical experience in FastAPI, Docker, and MLOps tools to turn machine learning models into real-world applications.

## Technical Skills

**Data Science & Modeling:** Python, PyTorch, Scikit-learn, TensorFlow/Keras, Pandas, NumPy, statistical analysis, model evaluation, Feature Engineering, recommendation systems, clustering
**Generative AI & NLP:** LLMs, RAG, LangChain, LlamaIndex, Sentence-Transformers, FAISS, ChromaDB, CLIP, Whisper, prompt engineering
**Engineering & MLOps:** FastAPI, Flask, REST APIs, Docker, Git, CI/CD, Redis, MLflow, model deployment, SQL, PostgreSQL, MongoDB, Supabase
**Languages:** Arabic (Native), French (Fluent), English (Professional)

## Professional Experience

**AI & MLOps Engineering Intern** – eGlobal Digital, Casablanca — Feb. 2026 – Jul. 2026

- Contributed to the development of **IVEP**, a production-grade multi-role SaaS platform featuring **30+ backend modules** and multiple AI services, built with FastAPI, MongoDB, Redis, Docker, and Next.js.
- Designed and evaluated a **hybrid recommendation system** combining behavioral signals, semantic representations, and time-aware popularity, using temporal data splitting for evaluation.
- Developed **RAG conversational services** using Sentence-Transformers, LangChain, and ChromaDB, and integrated multilingual AI pipelines with **Whisper and NLLB-200**.

**Full-Stack Developer Intern** – Big4Com, Marrakech — Apr. 2024 – Jun. 2024

- Developed **MyInvoice**, a full-stack invoicing platform for SMEs and freelancers, including customer management and automated PDF invoice generation.
- Designed and integrated **REST APIs with Node.js/Express.js** and developed a responsive **React.js** interface connected to business logic and persistent data management.

**Web Developer Intern** – Big4Com, Marrakech — Apr. 2023 – Jun. 2023

- Developed a web platform for a travel agency featuring **online booking, data management, and an administrative back office**.
- Designed the **relational database** and developed a responsive interface optimized for desktop and mobile devices.

## Projects

**WWW 2025 MM-CTR Challenge – Multimodal CTR Prediction** — GitHub

- Designed a cascaded multimodal pipeline on **3.6M training samples and 91,718 items**, using **CLIP** to extract and project 512D image/text representations into normalized 128D embeddings.
- Developed a CTR model combining a **Transformer, Deep Cross Network, and DNN**, achieving a **0.9495 validation AUC**, trained with PyTorch on **2× Tesla T4 GPUs**.

**Course Materials RAG – Multilingual AI Assistant** — GitHub

- Developed a multilingual **RAG system** for PDFs, text documents, and YouTube transcripts, using 768D Sentence-Transformers embeddings, **FAISS + MMR** retrieval, and Llama 3.3 70B.
- Developed a FastAPI + React/TypeScript application supporting document ingestion, multi-course management, contextual search, and source citation.

**MovieRec – Adaptive Hybrid Recommendation System** — GitHub

- Developed an adaptive recommendation engine using **MovieLens 100K (943 users, 1,682 movies)**, combining collaborative filtering, TF-IDF similarity, and popularity, with explicit *cold-start* handling.
- Implemented **4 recommendation strategies based on user engagement** and integrated them into a Flask/React/MongoDB application with authentication and the TMDB API.

## Education

**Master's Degree – EIDD, Data Exploration and Decision Support**, Université Sorbonne Paris Nord, Paris — 2025 – 2026
**Master's Degree – WISD, Web Intelligence and Data Science**, Université Sidi Mohamed Ben Abdellah, Fès — 2024 – 2026
**Professional Bachelor's Degree – Computer Systems and Software Engineering**, École Supérieure de Technologie, Essaouira — 2023 – 2024
**University Diploma of Technology – Computer Engineering**, École Supérieure de Technologie, Essaouira — 2021 – 2023

## Certifications

**Machine Learning in Python** · **Deep Learning with TensorFlow 2** · **Power BI** · **Data Ingestion with Pandas** — 365 Data Science, 2025